A completely specified nine-positive consistency equation

Nine positive inputs; accepted evaluator: 413 operations, degree 1,316,546. All fixed data below.

The only unknowns are $b, g, h, s, w, C, i, j, n \in \mathbb{Z}_{>0}$. All other letters are polynomial abbreviations or fixed constants. The equation is

$$\boxed{\left[\Theta^2 + \Omega^2(a_1 - I)\right]^2 - 4\Omega^2 a_1 \Theta^2 = 0.}$$

Source and ratio polynomials. For any displayed finite list put $p(e_1, \dots, e_t) = \sum_{\nu=1}^{t} B^{e_\nu}$. Define

$$\begin{gathered}
B = 2^{864026} b^2, \quad N_0 = B^{116}, \quad \ell = p(\mathcal{W}), \quad Z = p(\mathcal{T}), \\
u = \ell + g, \quad G_c = D_2(B)u^2 + D_1(B)u + D_0(B), \\
S_c = g + N_0(Z + 2G_c), \\
T_c = N_0 - 1 - \ell(b-1) + N_0 Z(B-2), \\
L_c = bB^{2197}, \quad R_+ = L_c(S_c + T_c + 1) - S_c, \quad R = R_+ - 1, \\
N = 2^{6480184} B^{1160}, \quad c_* = bB^{115}, \\
U = N^2 w, \quad Y = N^2 s, \quad M = RY, \quad P = 2M^2 U, \\
k = R_+ + h(P-1), \quad A = M(U+1), \quad B_1 = R + R_+, \quad W_1 = bw, \\
J_r = 4A - 5, \quad Q_r = 3W_1 C - 2(W_1^2 - 1), \\
m = k^2(4c_* - g) - 16c_*(C - kY)^2, \quad \Omega = J_r m, \\
a_0 = 1 + (Pk)^2 - k^2, \quad e = iC^2, \quad d_e = a_0 e, \quad V_0 = A d_e, \\
F_p = 1 + e(AV_0 - d_e), \quad G_p = A(F_p - 1) + F_p, \\
H = C + F_p(B_1 + jC), \quad V_1 = G_p H, \quad I = 1 + V_1^2 - H^2, \\
a_1 = a_0 F_p, \quad E_0 = J_r n + Q_r, \quad \Theta = E_0 - \Omega(V_0 + V_1).
\end{gathered}$$

The complete fixed lists, with $p(\mathcal{W})$ and $p(\mathcal{T})$ meaning the sum over all entries, are

$$\begin{aligned}
\mathcal{W} = (&0, 6, 60, 35, 15, 84, 19, 66, 42, 10, 33, 12, 95, 79, 28, 49, \\
&1, 108, 109, 111, 115, 90, 52, 54, 26, 100, 91, 8, 106, 36, 59), \\
\mathcal{T} = (&26, 122, 323, 382, 587, 628, 641, 694, 844, 1071, 1080, \\
&1157, 1249, 1259, 1448, 1508, 1625, 1754, 1791, 1805, 2026, 2048).
\end{aligned}$$

All 85 kernel terms. Equal-coefficient powers are grouped only by the explicitly defined finite-list notation p:

$$\begin{aligned}
D_0(B) &= 2K_0 B^{122} - 2p(323, 382, 1448, 1625) - 6B^{628}, \\
D_1(B) &= 2p(25, 32, 476, 541, 569, 604, 1019, 1026) \\
&\quad + 2p(1046, 1066, 1143, 1233, 1353, 1357, 1369) \\
&\quad + 2p(1500, 1690, 1718, 1732, 1755, 1917, 1940) \\
&\quad - 2p(0, 492, 661, 753, 981, 1697, 1795) - 8B^{513} - 6B^{1022} \\
&\quad - 2K_0 B^{62} - 2(K_0 + 1)B^{116} + 2\beta(K_0 + 1)B^{122} \\
&\quad - 2(M_C + M_F)B^{844} - 4\beta B^{1259}, \\
D_2(B) &= -\beta p(0, 87, 661, 843, 1795) \\
&\quad - p(29, 83, 119, 135, 338, 377, 393, 758, 812, 845) \\
&\quad - p(1076, 1176, 1256, 1295, 1512, 1632, 1794, 1807, 1843) \\
&\quad - 2p(105, 398, 611, 858, 1452, 1542) \\
&\quad + 2p(299, 326, 1326, 1521) \\
&\quad + p(384, 400, 691, 837, 1823, 1850) - 2\beta M_F B^{844} \\
&\quad - 4B^{907} - 2\beta^2 B^{1259}.
\end{aligned}$$

The complete fixed coefficient tuple. The divisions and floors below define constants only; the rational intermediate defining K_0 is an integer.

$$\begin{gathered}
\rho = 2^{89}, \quad \mathcal{B} = \rho^{4831}, \quad \beta = \mathcal{B} - 1, \\
\lambda = 2^{29}, \quad \theta = 2^{42}, \quad \mathcal{Q} = 2^{40}, \\
G_t(x) = \sum_{r=0}^{t-1} x^r, \quad \mu = 510 + 2^{17} + 2^{26} + 2^{28} + \lambda(2^{44} - 1).
\end{gathered}$$

$$\begin{aligned}
\mathcal{S} = &\{0, \dots, 456\} \cup \{495, \dots, 504\} \cup \{507, 508\} \\
&\cup \{517, \dots, 537\} \cup \{539, \dots, 549\} \\
&\cup \{567, \dots, 583\} \cup \{1203, 1807\}.
\end{aligned}$$

$$\begin{aligned}
M_C &= \mathcal{B} - 1 - \sum\nolimits_{r \in \mathcal{S} \setminus \{1\}} \rho^r - 2\rho^{507}, \\
M_F &= \mathcal{B} - 1 + 4 + \mu(\rho^{2409} + \rho^{3015}) \\
&\quad + \rho^{528} G_{10}(\rho) + \rho^{585} G_{11}(\rho) + \rho^{1203} + \rho^{1807} \\
&\quad + (2^{87} - 1)\rho^{4823} G_4(\rho) + (2^{38} - 1)\rho^{4827}, \\
K_0 &= \rho^{33} + \rho^{68} + \rho^{664} + \rho^{1268} + \rho^{4877} \\
&\quad + (\rho^{2409} + \rho^{3015})\Phi(1/\rho),
\end{aligned}$$

$$\begin{aligned}
\Phi(x) = {} & O G_{457}(x) + \lambda \sum\nolimits_{a=0}^{456} \ell_a x^{r_a} \\
&- (255\,2^9 + 2^{27})x^{374} - (255\,2^{18} + 2^{27})x^4 \\
&+ \lambda\big[(x^{539} + 2^{21}x^{517})G_{11}(2x) \\
&\quad + (2^{11}x^{528} + 2^{32}x^{495})G_{10}(2x) \\
&\quad + \theta(x^{539} + x^{1203} + x^{1807})\big] \\
&+ \big[(2^9 + \lambda(2 + 2^{33}))x^{567} \\
&\quad + (2^{18} + \lambda(2^{22} + 2^{33}))x^{575}\big] G_8(2x) \\
&+ (2^{27} + \lambda(2^{11} + 2^{32}))x^{583}.
\end{aligned}$$

The ranks and labels. The following 457 fixed quadruples are $t_r = (w_r, n_r, e_r, s_r)$. From the fixed integer $\mathcal{A}$ below, for $0 \le r < 430$ put

$$\begin{gathered}
b_r = \lfloor \mathcal{A}/1024^r \rfloor \bmod 1024, \quad u_r = b_r \bmod 256, \\
v_r = \lfloor b_r/256 \rfloor \bmod 2, \quad d_r = \lfloor b_r/512 \rfloor, \\
t_{2+r} = ((1 - d_r)(512 + u_r), 512 + r, d_r(256 + u_r), v_r).
\end{gathered}$$

Here $a \bmod b = a - b\lfloor a/b \rfloor$. The remaining quadruples are

index	range	t_r
0		$(2, 3, 3, 1)$
1		$(5, 0, 0, 0)$
$432 + r$	$0 \le r < 2$	$(0, 942 + r, 0, 942 + r)$
$434 + r$	$0 \le r < 2$	$(0, r, 0, r)$
$436 + r$	$0 \le r < 4$	$(0, (r \bmod 2) + 942\lfloor r/2 \rfloor, 0, 6)$
440		$(2, 2, 2, 0)$
441		$(3, 4, 4, 698)$
442		$(4, 4, 4, 0)$
$443 + r$	$0 \le r < 3$	$(1, 6, 1, 2 + r)$
$446 + 3r + t$	$0 \le r, t < 3$	$(r^2, 5, t, 5)$
455		$(256, 0, 0, 512)$
456		$(0, 0, 512, 512)$

For these bound indices define

$$\begin{aligned}
&\kappa_r = 2^{30} w_r + 2^{20} n_r + 2^{10} e_r + s_r, \\
&\ell_r = 2w_r + 2^{11} n_r + 2^{22} e_r + 2^{32} s_r, \\
&r_0 = 0, \quad r_1 = 1, \\
&r_t = 2 + \sum\nolimits_{r=2}^{456} \left\lfloor \frac{\mathcal{Q} + \kappa_t - \kappa_r - 1}{\mathcal{Q}} \right\rfloor \quad (2 \le t \le 456), \\
&O = 1 + 255\,2^9 + 255\,2^{18} + 2^{27} + \lambda.
\end{aligned}$$

Subscripts in the constant recipes are bound indices, not extra unknowns.

The integer $\mathcal{A}$. Digit values are $0, \dots, 61$ in the alphabet `0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz`

For the following 722 characters put $\mathcal{A} = \sum_{r=0}^{721} \epsilon_r 62^{721-r}$, where ϵ_r is the value of character r, starting at zero:

```
RCtYkANMQ1gmPqxuKhLmpi1mjCvu243YjyAxv9hbC2UsurRLykLFhuHVBGGUZ5
whVShV0FQcVVQSR2GS6JAW1DGYxGKlA4rJavFCW5i63LLSs8xEGvWTbLjSFULV
ODe48Kw2ghEaa5Q9PZZVHtki7C45jfVYROOzvVT9DAGjLOBlBSHQU8WPUF3SpC
kTfDQF7mUboMEAeyPWARCyClpnxtqgsit318hQgiIdKauxc6memnYNtycB0s7z
K8dMgYaLRxNjlcLvJzZ9Z5wdCG3wjd5P6ZVqC2SXVaimqXrTY5dNxbCH2B5VKA
Sp16kht0JV65p4VclYsI6zi6nIr7elpUCOX7IHdu6qKiu6Kt5Mq1YZjChEzG6y
W1WsPdMB71PL3oOCFqFncCtVEMUounam1fyhduQYoTyTCIEMqcD2BPQfMVaBa9
BZHQ8k7rpS3eiM7Rsak5uUsoP1Tf1uhWshndLuN8yz7UdD5QXM6AJ3ovPdWsqO
HGBeRaR8kch4R9JVbkAxS2ByaeE2O6czIT0sWtpasWDdZB9FuDQUFYhSvPPPr8
nSQmnMkF90CBzWkpeP81azGOWtiT2XTxpaLcR4ecaK1uZyvBleYknk6QTwECl5
XJ3nJBoP4sMTGaV6unoGzq9s1KN4KtXrVRuTpi6EuqizW9njcp9Uikq1VNjVtw
W1AGLJu3eaWgOEEF7vROqLuNbzEheYnyi4SORWVp
```

Absence of positive integer solutions is equivalent to Con(ZFC) by the inherited source/native calibration and the ordinary saturated Pell argument. This page specifies the equation; it does not supply an end-to-end formal Lean proof.